Myanmar Syllable Handwritten Collector

A system for collecting handwritten Myanmar syllable data, available in two versions: a modified PyQt5 desktop application and a touch-friendly web application. This project is developed as part of the **AI Engineering Fundamental Class Assignment**, based on the original implementation by **Dr. Ye Kyaw Thu, LU Lab., Myanmar**.

Overview

The original system provides a desktop interface for collecting handwritten Myanmar syllables using mouse or trackpad input. This project extends it in two directions: a **modified desktop version** with keyboard-driven workflow improvements, and a **web-based reimplementation** that brings data collection to touchscreen devices.

Key Goals:

- Enable handwriting data collection on **touchscreen devices** (phones, tablets)
- Reduce friction in the collection process with **keyboard shortcuts** and **auto-advance**
- Integrate dataset browsing and image generation into a **single interface**
- Preserve the original **data format** for full compatibility across both versions

Feature	**Original (Desktop)**	**Modified Desktop**	**Web**
Input Method	Mouse / Trackpad	Mouse / Trackpad + Keyboard	Touch + Mouse
Navigation	Manual (click)	Manual (click or A / D keys)	Auto (3 saves per line)
Undo	Not available	Not available	Available
Delete Samples	Not available	Not available	Available
Image Generation	CLI (3 commands)	CLI (3 commands)	Integrated (1 click)
Dataset Browser	Separate app	Separate app (bug-fixed)	Built-in
Accessibility	Desktop only	Desktop only	All devices on Browsers

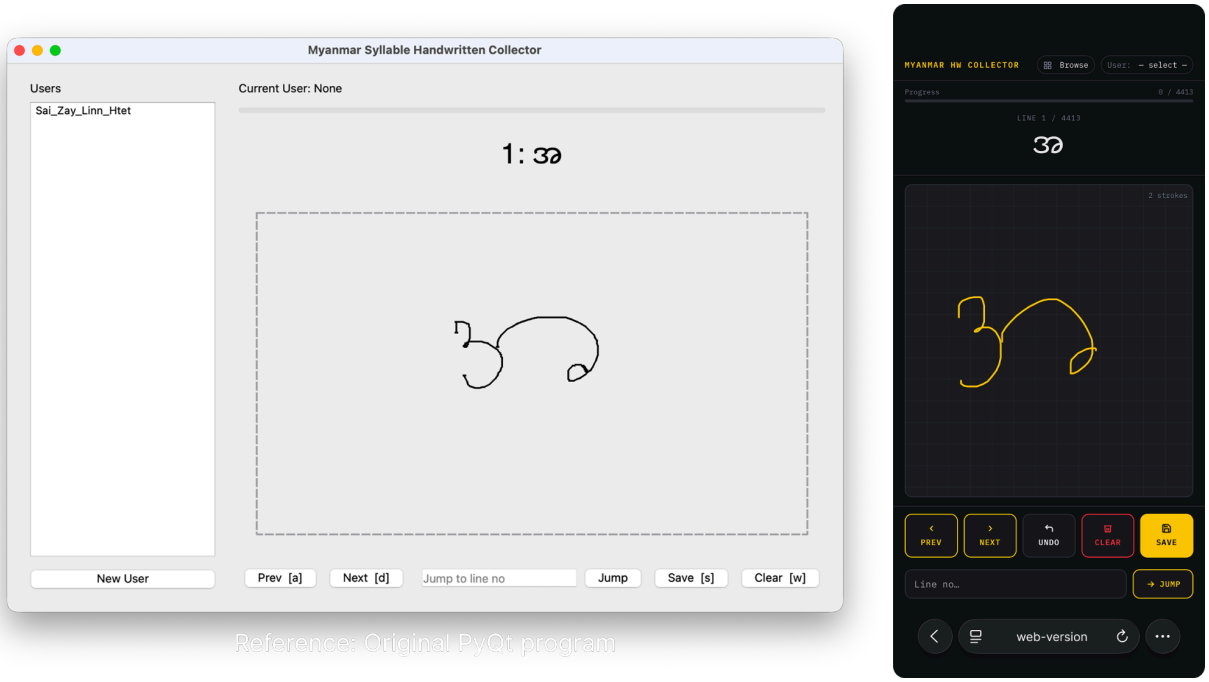

Figure 1. Sample UI for both modified desktop version and new web version

Prerequisites

Before running either the Desktop or Web version, ensure your environment is properly set up.

Python

Make sure Python is installed on your system.

You can verify this by running:

```
python --version
```

Virtual Environment Setup

It is recommended to use a virtual environment to manage dependencies.

1. Create a virtual environment

```
python -m venv venv
```

2. Activate the virtual environment

macOS / Linux:

```
source venv/bin/activate
```

Windows:

```
venv\Scripts\activate
```

3. Install dependencies

```
pip install -r requirements.txt
```

1. Modified Desktop (PyQt5)

Modified from the original by Dr. Ye Kyaw Thu. Changes focus on keyboard ergonomics and stability during long collection sessions.

New Features

Keyboard Shortcuts

Navigate and save without touching the mouse. Key bindings are shown in each button label.

Key	Action
`A`	Previous line
`D`	Next line
`S`	Save sample
`W`	Clear canvas
`Space`	Click focused button

Bug Fix - Dataset Browser Crash

`dataset_browser.py`: Added a directory check in `load_users()` to skip stray files in the dataset folder, preventing crashes on load.

```
 # Original - crashes if any non-directory file exists in dataset/
(e.g: .DS_Store file in MacOS)
for user in sorted(os.listdir(self.dataset_dir)):
    self.user_list.addItem(user)
```

```
# Fixed
for user in sorted(os.listdir(self.dataset_dir)):
    user_path = os.path.join(self.dataset_dir, user)
    if not os.path.isdir(user_path):
        continue
    self.user_list.addItem(user)
```

Quick Start (Desktop)

Ensure all steps in the **Prerequisites** section are completed.

1. Collect handwriting samples:

```
python hw_collector.py --file syl.txt
```

2. Browse the dataset:

```
python dataset_browser.py --dataset dataset --textfile syl.txt
```

3. Convert strokes to images:

Black stroke rendering -

```
python convert2image.py --dataset dataset --output single
--color_mode single
```

Different color per stroke -

```
 python convert2image.py --dataset dataset --output stroke
```

```
--color_mode stroke
```

Color progression by drawing order -

```
python convert2image.py --dataset dataset --output time --color_mode
time
```

2. Web App (Flask API)

A full reimplementation as a mobile-first web application. Preserves the original data format and collection logic while introducing new capabilities for touch devices and streamlined workflows.

New Features

Touch-Based Canvas with Stroke Count

Draw directly with fingers or mouse. Stroke count is displayed live on the canvas.

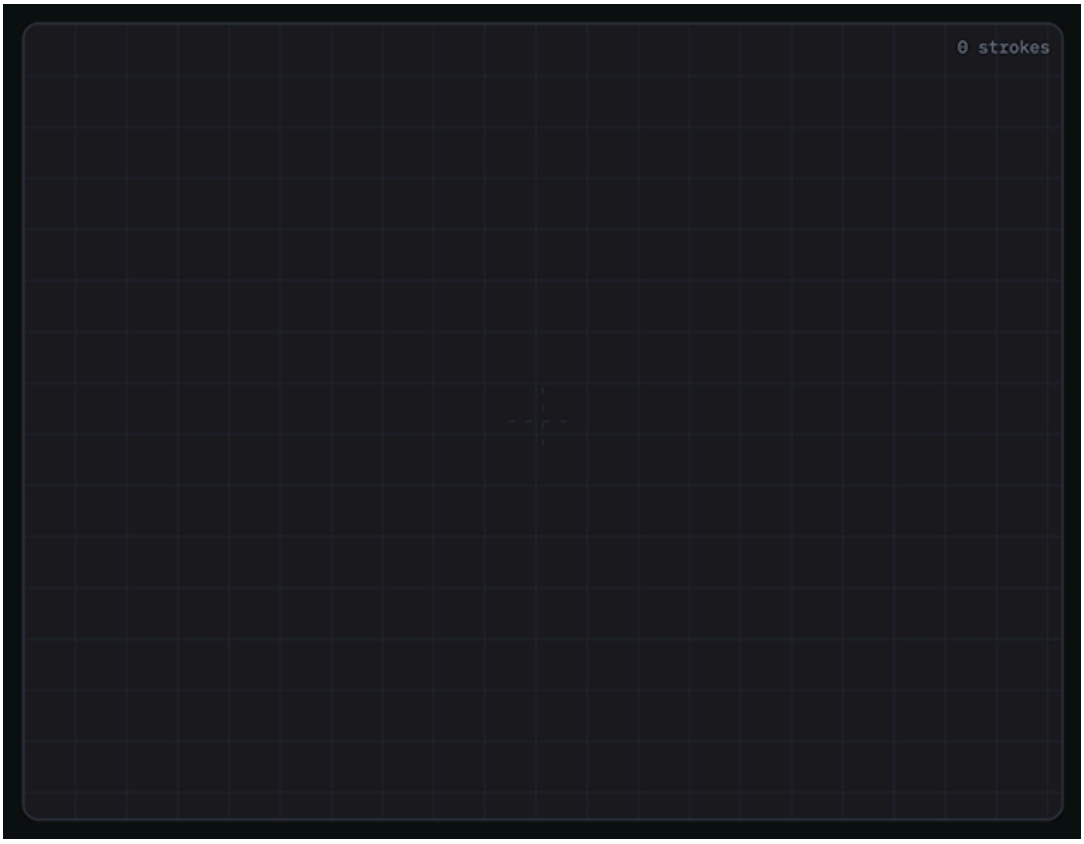

Figure 2. Touch screen canvas with stroke count

Guided Collection - Auto-Advance (3 Saves per Line)

Each syllable requires 3 samples before automatically moving to the next line, enforcing consistency without manual navigation.

Figure 3. Toast UI

Undo and Extended Canvas Controls

- Undo last stroke
- Clear canvas
- Navigate between lines
- Jump to a specific line number

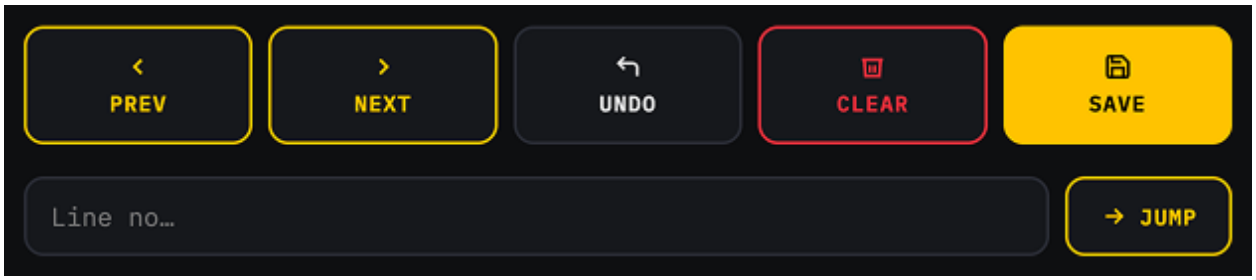

Figure 4. Canvas Controls

Dataset Management and Progress Tracking

Delete incorrect samples and track per-user progress directly in the interface.

Figure 5. Progress Bar

Integrated Dataset Browser with Real-Time Image Generation

Browse all collected samples in a visual grid. Images are generated on demand in 3 modes, no separate CLI commands needed.

- **Single Mode**: black stroke rendering
- **Stroke Mode**: different color per stroke
- **Time Mode**: color progression by drawing order

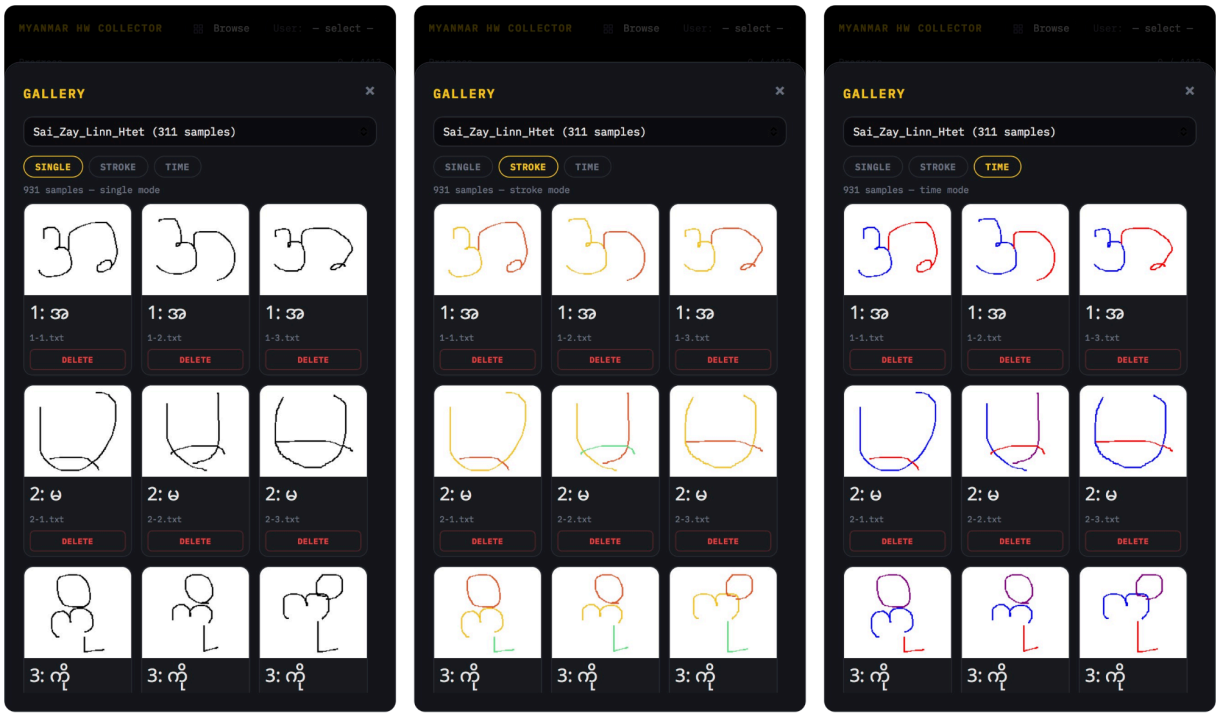

Figure 6. Build-in dataset browser and one-click preview

Quick Start (Web)

Ensure all steps in the **Prerequisites** section are completed.

1. Start the server:

```
python hw_server.py --file syl.txt --port 5005
```

2. Open in browser:

```
# On your computer
http://localhost:5005
# On phone / tablet (same Wi-Fi network)
http://<YOUR_IP_ADDRESS>:5005
```

Technical Notes

- Stroke normalization and rendering logic are adapted from the original `convert2image.py`
- Dataset format (`.txt` stroke files) is unchanged across both versions for full compatibility
- Web image generation is performed lazily - only on request, with caching
- Web frontend uses **vanilla JavaScript and HTML5 Canvas** (no framework dependencies)
- Web backend uses **Flask + Pillow**
- Desktop keyboard input requires `Qt.StrongFocus` to be set on `MainWindow` so events are not consumed by child widgets

Dataset Structure

```
dataset/
  └── <user_name>/
        ├── user_info.json
        ├── 1-1.txt
        ├── 1-2.txt
        └── ...
single/
stroke/
time/
  └── <user_name>/
        └── *.png
```

Core API Endpoints

Method	Endpoint	Description
`GET`	`/api/lines`	Load syllable list
`GET`	`/api/users`	List users and progress
`POST`	`/api/users`	Create new user
`POST`	`/api/save`	Save stroke data
`POST`	`/api/gallery/<user>/<mode>`	Load dataset images (cached)
`DELETE`	`/api/delete/<user>/<file>`	Delete sample and associated images

Attribution

This project is built upon the original work by:

Dr. Ye Kyaw Thu
LU Lab., Myanmar - AI Engineering Fundamental Class

Original components:

- `hw_collector.py` handwriting collection main app
- `dataset_browser.py` dataset browsing tool
- `convert2image.py` stroke-to-image conversion
- `syl.txt` syllable list

The core logic and dataset format are preserved. This project extends the original by introducing keyboard ergonomics (Version 1) and a full web-based interface with additional functionality (Version 2).

Development Notes

This project was developed using a combination of personal implementation and modern development tools, including AI-assisted coding workflows. All architectural decisions, modifications, and integrations were designed and validated by the author.

Purpose

This project is intended for:

- Educational use in NLP / AI engineering courses
- Research in Myanmar handwriting data collection
- Extension and experimentation by students